CSE400 – Fundamentals of Probability in Computing
Lecture 4: Joint Probability and Conditional Probability

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Lecture - 4

1. Engineering Applications of Probability

Examples of engineering systems where probability is applied include:

Speech Recognition System

A vocabulary such as Hello, Yes, No, Bye can be represented using waveform templates. Different speakers (male, female, child) produce different waveform patterns for the same word.

Radar System

Radar signals are transmitted and reflected from objects such as aircraft. The received signal is processed to determine object detection.

Communication Network

Data is transmitted from a source to a destination through multiple network nodes and devices.

2. Introduction to Probability Theory

2.1 Experiments, Sample Space, and Events

Definition: Experiment (E)

An experiment is a procedure that produces some result.

Example:

Tossing a coin five times E_5.

Definition: Outcome (ξ)

An outcome is a possible result of an experiment.

Example:

One outcome of E_5:

$\xi_1 = HHTHT$

Definition: Event

An event is a set of outcomes of an experiment.

Example:

Let event C in experiment E_5 be:

$C = \{$all outcomes consisting of an even number of heads$\}$

Definition: Sample Space (S)

The sample space is the collection or set of all possible distinct outcomes of an experiment.

These outcomes are:

Mutually Exclusive

Example: In a coin flip, you can obtain heads or tails, but not both.

Collectively Exhaustive

All possible outcomes are included in the sample space.

Example: Flip a coin.

Properties of Sample Space

S is the universal set of outcomes of an experiment.

The sample space can be:

Discrete

Countably infinite

Continuous

Examples of Experiments

Flipping a fair coin once

Rolling a cubical die with numbered faces

Rolling two dice

Flipping a coin until a tails occurs

Random number generator with interval $[0, 1]$

3. Axioms of Probability

Definition: Probability

Probability is a measure of the likelihood of events or a function of an event that produces a numerical quantity measuring the likelihood of that event.

Axioms

The following statements are taken as self-evident and require no proof.

For any event A:

$$0 \leq Pr(A) \leq 1$$

If S is the sample space:

$$Pr(S) = 1$$

If A and B are mutually exclusive ($A \cap B = \emptyset$):

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For mutually exclusive sets A_i:

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

4. Corollary from Probability Axioms

Let A be the null event for all values of i greater than n.

Corollary 2.1

For M mutually exclusive events:

$$Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} Pr(A_i)$$

Axiom 3 is equivalent to Corollary 2.1 when the sample space is finite.

The generality of Axiom 3 is necessary when the sample space contains infinitely many points.

5. Propositions from Probability Axioms

Proposition 2.1

$$Pr(A^c) = 1 - Pr(A)$$

Proposition 2.2

If $A \subset B$:

$$Pr(A) \leq Pr(B)$$

Proposition 2.3

For any sets A and B:

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Proposition 2.4 (Inclusion–Exclusion Principle)

$$Pr(A_1 \cup A_2 \cup ... \cup A_M) = \sum_{i=1}^{M} Pr(A_i) - \sum_{i_1 < i_2} Pr(A_{i_1} A_{i_2}) + ... + (-1)^{M+1} Pr(A_1 A_2 ... A_M)$$

6. Assigning Probabilities

Classical Approach

Example 2.6

Coin flipping experiment:

Compute

$Pr(H), Pr(T)$

Example 2.7

Dice rolling experiment:

Compute

$Pr(even\ number\ is\ rolled)$

Example 2.8

Two dice are rolled.

Let event A:

Sum of the two dice equals five.

Compute:

$Pr(A)$

Relative Frequency Approach

Example: Dice experiment where event A = sum of two dice equals five.

Simulation results:

n	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000
n_A	96	200	314	408	521	630	751	859	970	1095
n_A/n	0.096	0.100	0.105	0.102	0.104	0.105	0.107	0.107	0.108	0.110

Inference

To obtain the exact probability of an event, the experiment must be repeated infinitely many times.

Many random phenomena of interest are not repeatable.

7. Joint Probability

Motivation

Events are not always mutually exclusive.

We are interested in the probability of the intersection of events A and B.

Notation

Joint probability is denoted as:

$$Pr(A, B) \quad \text{or} \quad Pr(A \cap B)$$

Multiple Events

For events $A_1, A_2, ..., A_M$:

$$Pr(A_1, A_2, ..., A_M)$$

8. Example 1: Card Deck

Consider a deck of 52 playing cards.

Define events:

A: Red card selected

B: Number card selected (Ace counted as number)

C: Heart card selected

Find:

$Pr(A), Pr(B), Pr(C), Pr(A, B), Pr(A, C), Pr(B, C)$

Solution

Individual Probabilities

$$Pr(A) = \frac{26}{52} = \frac{1}{2}$$

$$Pr(B) = \frac{40}{52} = \frac{5}{26}$$

$$Pr(C) = \frac{13}{52} = \frac{1}{4}$$

Joint Probabilities

$$Pr(A, B) = \frac{20}{52} = \frac{5}{13}$$

$$Pr(A, C) = \frac{13}{52} = \frac{1}{4}$$

$$Pr(B, C) = \frac{10}{52}$$

9. Example 2: Costume Party

Alex has the following clothing:

Tops:

3 white T-shirts

1 radioactive green cape

Bottoms:

2 flannel dinosaur pajama pants

4 polka-dot boxer shorts

Step 1: Analyze Tops

Total tops:

$$3 + 1 = 4$$

$$Pr(Cape) = \frac{1}{4}$$

Step 2: Analyze Bottoms

Total bottoms:

$$2 + 4 = 6$$

$$Pr(Boxers) = \frac{4}{6} = \frac{2}{3}$$

Step 3: Joint Probability

$$Pr(Cape \cap Boxers) = Pr(Cape) \times Pr(Boxers)$$

$$= \frac{1}{4} \times \frac{4}{6} = \frac{1}{6}$$

Correct Option: C

10. Conditional Probability

Motivation

The occurrence of one event may depend on another event.

Definition

The probability of event A given event B:

$$Pr(A|B) = \frac{Pr(A,B)}{Pr(B)}$$

where

$$Pr(B) > 0$$

Product Rule

$$Pr(A, B) = Pr(A|B)Pr(B) = Pr(B|A)Pr(A)$$

11. Example 3: Cards Without Replacement

Two cards are selected without replacement.

Define:

A: First card is a spade

B: Second card is a spade

Find:

$Pr(B|A)$

Initial State

Total cards = 52

Spades = 13

After Event A

One spade removed.

Remaining:

Total cards = $52 - 1 = 51$

Spades = $13 - 1 = 12$

Calculation

$$Pr(B|A) = \frac{12}{51}$$

12. Example 4: Game of Poker

Five cards are dealt.

A flush means all cards are from the same suit.

Part 1: Flush in Spades

$$P(1st\ spade) = \frac{13}{52}$$

$$P(2nd\ spade) = \frac{12}{51}$$

$$P(3rd\ spade) = \frac{11}{50}$$

$$P(4th\ spade) = \frac{10}{49}$$

$$P(5th\ spade) = \frac{9}{48}$$

$$P(Spade\ Flush) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Part 2: Any Flush

There are four suits:

Spades, Hearts, Diamonds, Clubs.

Since events are mutually exclusive:

$$P(Any\ Flush) = 4 \times P(Spade\ Flush)$$

13. Example 5: The Missing Key

Bob is 80% certain the key is in his jacket.

Left pocket:

$$P(L) = 0.4$$

Right pocket:

$$P(R) = 0.4$$

In jacket:

$$P(K) = 0.8$$

Goal:

$$P(R|L^c)$$

Probability that the key is in the right pocket given it is not in the left pocket.

Calculation

$$P(R|L^c) = \frac{P(R \cap L^c)}{P(L^c)}$$

Since R implies L^c:

$$P(R \cap L^c) = P(R)$$

$$P(L^c) = 1 - P(L)$$

Therefore

$$P(R|L^c) = \frac{0.4}{1-0.4} = \frac{0.4}{0.6} = \frac{2}{3}$$

Correct Option: C